Examiner only

13. *(a)* During the 2016 Olympic games, the diving pool had to be closed after the water turned green. A report in the media incorrectly suggested that there was too much chlorine in the pool.

When chlorine gas dissolves in cold water, a pale green solution is formed. In this solution, the following equilibrium is established.

$$\underset{\text{pale green}}{Cl_2(aq)} + H_2O(l) \rightleftharpoons H^+(aq) + Cl^-(aq) + \underset{\text{colourless}}{HClO(aq)}$$

(i) Chemical equilibria are often described as dynamic equilibria.

Explain the term *dynamic equilibrium*. [1]

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(ii) Use Le Chatelier's principle to explain why the pale green colour disappears if sodium hydroxide solution is added. [2]

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(b) Chlorine can react directly with metals to form chlorides.

(i) I. Calcium chloride can exist as an anhydrous salt or as a hydrated salt, $CaCl_2.xH_2O$.

In an experiment to determine the extent of hydration a sample of hydrated calcium chloride, $CaCl_2.xH_2O$, with a mass of 3.29 g was heated to remove all water of crystallisation. The solid remaining had a mass of 1.67 g.

Calculate the value of x in the formula $CaCl_2.xH_2O$.

You **must** show your working. [3]

x = ...

II. Suggest how a student doing this experiment would ensure that all the water had been removed. [1]

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(ii) The melting temperature of sodium chloride is 1074 K but sodium iodide has a melting temperature of 934 K.

Suggest why the melting temperature of sodium iodide is lower than that of sodium chloride. [1]

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(c) Chlorine forms molecules and ions with other halogens.

(i) While chlorine has a boiling temperature of 238 K, the boiling temperature of iodine monochloride, ICl, is 371 K.

Suggest why the boiling temperature of iodine monochloride is higher than that of chlorine. [2]

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(ii) Name the shape of the $[ClF_6]^+$ ion. [1]

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(iii) $[ClF_2]^+$ and $[ClF_2]^-$ are two other ions containing a chlorine atom.

A student said that their shapes must be different.

Is he correct? Justify your answer using VSEPR theory. [3]

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END OF PAPER